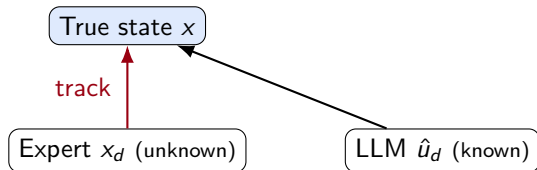


Certified Tracking of LLM-Predicted Trajectories

What Is Tracking What, Using What

Three trajectories in play:

- True state $x(t)$ — actual robot, **known dynamics**
- Desired/expert state $x_d(t)$ — from RRT/MPC, **never observed at test time**
- LLM prediction $\hat{u}_d$ — estimate of expert input u_d , **computable at test time**



Goal: design feedback so x provably tracks the *unobservable* x_d , using only $\hat{u}_d$.

Known Quantities

Design-time (always known)

- True dynamics structure: $\dot{x} = f(x, u)$ (or its linearization A, B)
- Freedom to design a feedback gain / metric (K , or $M(x), \alpha$)
- Calibration data $\{(x_{d,i}, u_{d,i}, \hat{u}_{d,i}, E_i)\}_{i=1}^N$, exchangeable with test conditions

Test-time (observed online)

- Environment E
- LLM prediction $\hat{u}_d = u_{vlm}(x, E)$
- True state $x(t)$ (measured)

Unknown Quantities & the Core Definitions

Unknown at test time

- Expert trajectory and control $x_d(t), u_d(t)$ — RRT/MPC is *not* re-run online
- Realized prediction gaps d — only its *calibrated statistics* are known

Tracking error:

$$e := x - x_d$$

Implementable controller:

$$u := -K\hat{e} + \hat{u}_d, \quad \hat{e} := x - \hat{x}_d$$

Where the two tools split the work:

- **Contraction / Lyapunov** → guarantees e stays bounded, *given* $\|d\|$ is bounded
- **Conformal prediction** → supplies that bound on $\|d\|$, from calibration data alone

Error Dynamics With the Implementable Controller

Substitute $\hat{e} = x - \hat{x}_d$ **into** $e = x - x_d$:

$$\hat{e} = (x - x_d) + (x_d - \hat{x}_d) = e - d_x, \quad d_x := \hat{x}_d - x_d$$

Error dynamics (same steps as before, now with $u = -K\hat{e} + \hat{u}_d = -K(e - d_x) + \hat{u}_d$):

$$\dot{e} = Ae + B(-K(e - d_x) + \hat{u}_d - u_d) = (A - BK)e + B \underbrace{(\hat{u}_d - u_d)}_{d_u} + BK d_x$$

Two disturbance terms now, not one

$$\dot{e} = (A - BK)e + Bd_u + BKd_x$$

$d_u = \hat{u}_d - u_d$ (LLM control-prediction gap) and $d_x = \hat{x}_d - x_d$ (LLM state-prediction gap) — **both** unobservable at test time, **both** need conformal bounds.

Conformal Prediction: Two Separate Nonconformity Scores

Calibration data: N past triples, assumed **exchangeable** with the test environment:

$$\{(x_{d,i}, u_{d,i}, \hat{u}_{d,i}, E_i)\}_{i=1}^N$$

Score 1 — control-prediction gap:

$$S_i^u := \sup_{t \in [0, T]} \|\hat{u}_d(x_{d,i}, E_i) - u_{d,i}(t)\|$$

Score 2 — state-prediction gap:

$$S_i^x := \sup_{t \in [0, T]} \|\hat{x}_d(E_i) - x_{d,i}(t)\|$$

Two calibrated quantiles, at confidence levels $1 - \delta_1$ and $1 - \delta_2$:

$$q_{1-\delta_1}^u = \lceil (1 - \delta_1)(N + 1) \rceil\text{-th smallest of } S_1^u, \dots, S_N^u$$

$$q_{1-\delta_2}^x = \lceil (1 - \delta_2)(N + 1) \rceil\text{-th smallest of } S_1^x, \dots, S_N^x$$

Each score has its own probabilistic guarantee: $\mathbb{P}[\|d_u\| \leq q_{1-\delta_1}^u] \geq 1 - \delta_1$, $\mathbb{P}[\|d_x\| \leq q_{1-\delta_2}^x] \geq 1 - \delta_2$ — separately.

Combining the Two Bounds: Boole's Inequality

Boole's inequality (union bound — holds for *any* two events, no independence needed):

$$\mathbb{P}[F_1 \cup F_2] \leq \mathbb{P}[F_1] + \mathbb{P}[F_2] \leq \delta_1 + \delta_2$$

Since $\mathbb{P}[F_1^c \cap F_2^c] = 1 - \mathbb{P}[F_1 \cup F_2]$:

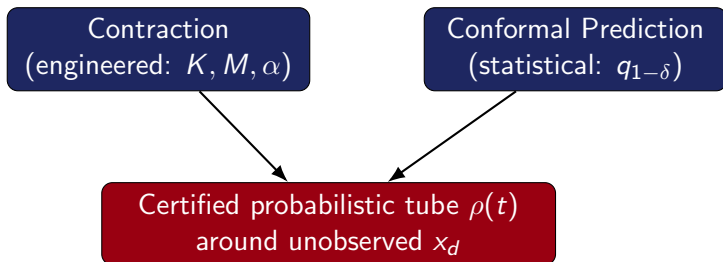
Combined probabilistic guarantee

$$\mathbb{P}[\|d_u\| \leq q_{1-\delta_1}^u \text{ and } \|d_x\| \leq q_{1-\delta_2}^x] \geq 1 - \delta_1 - \delta_2 =: 1 - \delta$$

Standard choice : split the overall budget evenly, $\delta_1 = \delta_2 = \delta/2$.

The Resulting Certificate (Schematic)

$$\mathbb{P}\left[\|x(t) - x_d(t)\| \leq \rho(t) \quad \forall t \in [0, T] \right] \geq 1 - \delta$$



Next: instantiate this on a linear system (Lyapunov), then a nonlinear one (contraction).

Linear Dynamics Setup

True dynamics (known linear model):

$$\dot{x} = Ax + Bu$$

Expert/desired trajectory:

$$\dot{x}_d = Ax_d + Bu_d$$

Implementable controller (feedback on $\hat{e}$, *not* the unobservable e):

$$u = -K\hat{e} + \hat{u}_d, \quad \hat{e} := x - \hat{x}_d, \quad e := x - x_d$$

Error dynamics (derived earlier from $\hat{e} = e - d_x$):

$$\dot{e} = (A - BK)e + B \underbrace{(\hat{u}_d - u_d)}_{d_u} + BK \underbrace{(\hat{x}_d - x_d)}_{d_x}$$

Lyapunov Candidate & Contraction Condition

Let $P \succ 0$ and define

$$V(e) = e^\top P e$$

Design requirement (Lyapunov equation, guarantees $A - BK$ Hurwitz):

$$(A - BK)^\top P + P(A - BK) \preceq -2\alpha P, \quad \alpha > 0$$

For linear system contraction is lyapunov for error dynamics with $M = P$

Differentiating V Along Trajectories

Product rule, then substitute $\dot{e} = (A - BK)e + Bd_u + BKd_x$ — **two** disturbance terms produce **two** cross terms:

$$\dot{V} = \dot{e}^\top P e + e^\top P \dot{e} = e^\top [(A - BK)^\top P + P(A - BK)] e + 2e^\top P B d_u + 2e^\top P B K d_x$$

Using the design condition:

$$\dot{V} \leq -2\alpha V + 2e^\top P B d_u + 2e^\top P B K d_x$$

Next: bound each cross term separately using only $\|d_u\|$, $\|d_x\|$ and $\sqrt{V}$ — not their directions.

Bounding Both Cross Terms via Cauchy–Schwarz

$P = P^{1/2}P^{1/2}$, then Cauchy–Schwarz $p^\top q \leq \|p\| \|q\|$, then bound each factor):

$$2e^\top PBd_u = 2(P^{1/2}e)^\top (P^{1/2}Bd_u) \leq 2\|P^{1/2}e\| \|P^{1/2}Bd_u\|$$

$$2e^\top PBKd_x = 2(P^{1/2}e)^\top (P^{1/2}BKd_x) \leq 2\|P^{1/2}e\| \|P^{1/2}BKd_x\|$$

Using $\|P^{1/2}e\| = \sqrt{V}$ and $\|P^{1/2}\| = \sqrt{\lambda_{\max}(P)}$ in both:

Result — two bounded cross terms

$$2e^\top PBd_u \leq 2\sqrt{\lambda_{\max}(P)} \|B\| \|d_u\| \sqrt{V}, \quad 2e^\top PBKd_x \leq 2\sqrt{\lambda_{\max}(P)} \|BK\| \|d_x\| \sqrt{V}$$

Cross Terms, Conformally Bounded

With $\|d_u\| \leq q_{1-\delta_1}^u$ and $\|d_x\| \leq q_{1-\delta_2}^x$ *simultaneously*, w.p. $\geq 1 - \delta$ (Boole's inequality, previous slides):

$$\dot{V} \leq -2\alpha V + \underbrace{2\sqrt{\lambda_{\max}(P)} \left(\|B\| q_{1-\delta_1}^u + \|BK\| q_{1-\delta_2}^x \right)}_{=:c} \sqrt{V}$$

Same shape as the single-disturbance case — the two LLM-prediction gaps simply *add* inside the constant c , each weighted by how strongly it enters the dynamics (B vs. BK).

Comparison Lemma $\Rightarrow$ Tracking Tube

Let $W = \sqrt{V}$. Then $\dot{W} \leq -\alpha W + \frac{c}{2}$, giving

$$W(t) \leq W(0)e^{-\alpha t} + \frac{c}{2\alpha}(1 - e^{-\alpha t})$$

Since $\sqrt{\lambda_{\min}(P)} \|e(t)\| \leq W(t)$:

$$\|e(t)\| \leq \sqrt{\frac{\lambda_{\max}(P)}{\lambda_{\min}(P)}} \|e(0)\| e^{-\alpha t} + \frac{\sqrt{\lambda_{\max}(P)} (\|B\| q_{1-\delta_1}^u + \|BK\| q_{1-\delta_2}^x)}{\alpha \sqrt{\lambda_{\min}(P)}} (1 - e^{-\alpha t})$$

Probabilistic guarantee

$$\mathbb{P} \left[\|x(t) - x_d(t)\| \leq \rho_{\text{lin}}(t) \quad \forall t \in [0, T] \right] \geq 1 - \delta$$

Exponential decay of initial error + a steady-state ball of radius $\propto (q_{1-\delta_1}^u + q_{1-\delta_2}^x)/\alpha$ —
without ever observing x_d or u_d , and using only the implementable controller.

Example: Planar Quadrotor Model

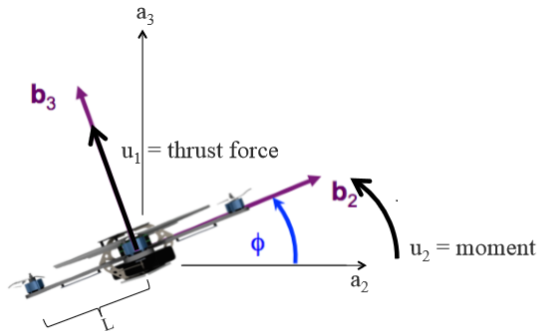


Figure 1: Planar quadrotor model and the coordinate systems.

State: $x = (y, z, \phi)$, Input: $u = (u_1, u_2)$

$$\begin{bmatrix} \ddot{y} \\ \ddot{z} \\ \ddot{\phi} \end{bmatrix} = \begin{bmatrix} 0 \\ -g \\ 0 \end{bmatrix} + \begin{bmatrix} -\frac{1}{m} \sin \phi & 0 \\ \frac{1}{m} \cos \phi & 0 \\ 0 & \frac{1}{J_{xx}} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

This is **nonlinear** in ϕ — no constant A, B describes it globally. The Lyapunov argument above does not directly apply.

Linearizing About Hover

Equilibrium: $\phi = 0$, $u_1 = mg$, $u_2 = 0$ (thrust balances gravity). Define deviations $\delta y, \delta z, \phi, \delta u_1 := u_1 - mg, \delta u_2 := u_2$, apply $\sin \phi \approx \phi$, $\cos \phi \approx 1$, drop second-order terms:

$$\ddot{y} \approx -g \phi, \quad \ddot{z} \approx \frac{1}{m} \delta u_1, \quad \ddot{\phi} \approx \frac{1}{J_{xx}} \delta u_2$$

State $x = (y, \dot{y}, z, \dot{z}, \phi, \dot{\phi}) \in \mathbb{R}^6$, **input** $u = (\delta u_1, \delta u_2)$:

$$\dot{x} = Ax + Bu, \quad A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -g & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{m} & 0 \\ 0 & 0 \\ 0 & \frac{1}{J_{xx}} \end{bmatrix}$$

Note the natural block structure: $\{z, \dot{z}\}$ is driven only by u_1 ; $\{y, \dot{y}, \phi, \dot{\phi}\}$ only by u_2 .

Designing K and Solving for P (z-Channel Example)

Subsystem: $x_z = (z, \dot{z})$, $A_z = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $B_z = \begin{bmatrix} 0 \\ 1/m \end{bmatrix}$.

Pole placement for closed-loop poles at $-2, -2.5$:

$$K_z = \begin{bmatrix} 5 & 4.5 \end{bmatrix}, \quad A_z - B_z K_z = \begin{bmatrix} 0 & 1 \\ -5 & -4.5 \end{bmatrix}, \quad \text{eig} = \{-2, -2.5\}$$

Solve the Lyapunov design condition for $\alpha = 1.8$, i.e. find $P \succ 0$ with

$$(A_z - B_z K_z)^\top P + P(A_z - B_z K_z) \preceq -2\alpha P$$

$$P = \begin{bmatrix} 128.69 & 46.43 \\ 46.43 & 17.38 \end{bmatrix}, \quad \lambda_{\min}(P) = 0.558, \quad \lambda_{\max}(P) = 145.51$$

Plugging (K_z, P, α) into $\rho_{\text{lin}}(t)$ from the Lyapunov slides — same applies to other subsystems $\{y, \dot{y}, \phi, \dot{\phi}\}$.

Designing M for the *Nonlinear* z-Channel

Keep the nonlinearity: apply $u_1 = mg - k_p e_z - k_d \dot{e}_z$ directly (no cancellation of $\cos \phi$).

Closed-loop z-error dynamics stay nonlinear:

$$\ddot{e}_z = (\cos \phi - 1)g - \frac{\cos \phi}{m}(k_p e_z + k_d \dot{e}_z), \quad \xi = (e_z, \dot{e}_z)$$

Jacobian (same $K_z = [5 \ 4.5]$):

$$\frac{\partial f_{cl}}{\partial \xi} = \begin{bmatrix} 0 & 1 \\ -\frac{k_p}{m} \cos \phi & -\frac{k_d}{m} \cos \phi \end{bmatrix}$$

Operating envelope $|\phi| \leq \phi_{\max} = 30^\circ$ ($\cos \phi_{\max} = 0.866$, worst case — smallest $\cos \phi$, weakest stiffness/damping):

$$\text{Find } M \succ 0 : \quad \frac{\partial f_{cl}}{\partial \xi}^\top M + M \frac{\partial f_{cl}}{\partial \xi} \preceq -2\alpha M \quad \forall |\phi| \leq \phi_{\max}$$

$$M = \begin{bmatrix} 5.26 & 1.33 \\ 1.33 & 0.63 \end{bmatrix}, \quad \alpha = 1.0, \quad \lambda_{\min}(M) = 0.277, \quad \lambda_{\max}(M) = 5.61$$

Designed at worst-case $\phi = \phi_{\max}$; holds for every $\phi \in [-30^\circ, 30^\circ]$ — unlike the linear P , which loses contraction past $|\phi| \approx 40^\circ$.

State-Dependent Contraction Condition for better notion of distance

Find $M(x, t) \succ 0$, $ml \preceq M \preceq \bar{m}l$, **such that:**

$$\dot{M} + M \frac{\partial f_{cl}}{\partial x} + \frac{\partial f_{cl}^\top}{\partial x} M \preceq -2\alpha M$$

Riemannian energy between x and x_d , along geodesic γ :

$$E(\gamma(\cdot, t)) = \int_0^1 \delta\gamma(s, t)^\top M(\gamma(s, t)) \delta\gamma(s, t) ds$$

Riemannian distance (replaces $\|e\|$ from the linear case):

$$d_M(x, x_d) = \sqrt{E(\gamma(\cdot, t))} = \|\Theta(\gamma(1, t)) \delta\gamma(1, t)\|, \quad M = \Theta^\top \Theta$$

This is exactly $\|e\|_P \rightarrow d_M(x, x_d)$ generalized from constant P to state-dependent $M(x)$.

Same Argument, Nonlinear Dress

Differentiate the energy along trajectories (mirrors $\dot{V}$ from before):

$$\dot{E}(\gamma(\cdot, t)) \leq -2\alpha E(\gamma(\cdot, t)) + 2\sqrt{m} d_M(x, x_d) \|d\|$$

where $d = \hat{u}_d - u_d$ is again bounded via conformal prediction: $\|d\| \leq q_{1-\delta}$ w.p. $\geq 1 - \delta$.

Same comparison-lemma machinery applies, replacing $\sqrt{V} \rightarrow d_M$:

$$d_M(x(t), x_d(t)) \leq d_M(x(0), x_d(0)) e^{-\alpha t} + \frac{\sqrt{m} q_{1-\delta}}{\alpha} (1 - e^{-\alpha t})$$